Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/15/2022
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rweber@matsdev.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
PI Address 1:	3090 N Daniels Ct		
PI Address 2:			
PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) program [1] to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Four prototype compositions have been levitated and melted on ISS. Measurements of thermophysical properties, density, thermal expansion coefficient, viscosity and surface tension were made over a wide temperature range. Samples, including glasses, were recovered for ground based analysis and comparison with materials made on Earth. The work underpins development of a materials property-structure data set that can help in development of new materials and in meeting requirements of material utilization and processing in space. Additional samples are being prepared for measurements on ISS in 2023. Work is being published to provide data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties and structure of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property and atomic structure data that can be used to optimize models used to design new materials from first principles.
Task Progress:	This project is part of a collaborative international Japan Aerospace Exploration Agency (JAXA) project to investigate: (i) properties of liquid phase and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS), and (ii) atomic structure, using ground-based synchrotron and neutron studies. We, the NASA investigators, are supplying samples for experiments using the ISS ELF to provide some of the first results from the instrument for US investigators. In addition to the scientific research, the proposed project will help to establish details of working with JAXA to provide flight samples and help to characterize the ELF capabilities by measuring US investigator materials. Samples were launched on SpX-20 in spring of 2020 and on SpX-22 in June 2021. Technical activities during the reporting period included: (i) Performing flight experiments on samples that were launched on SpX-22, (ii) analyzing data from flight experiments, (iii) performing ground-based measurements on samples recovered from the flight experiments, (iv) publishing/presenting results, and (v) preparing additional samples that will be launched in early 2023. Two sets of flight experiments have been performed. The first set was in August 2021; the second was in December 2021 through March 2022. A total of 26 samples covering 7 different compositions were investigated. For four compositions, multiple samples were stably levitated and fully melted. Drop size and oscillation power spectra were measured, and samples (glass for one composition) were recovered. Analysis was performed on the video images, drop oscillation power spectra, and temperature data to obtain the temperature-dependent properties of the liquids. During the early part of the work, considerable effort was put into developing procedures for the analysis and to enable precise synchronization of the temperature and other measured data. The video data were received as .TS video files. These were converted into a sequence of bitmap images (.BMP) using Adobe Premier Pro. The bitmap images were analyzed using a calibration sample of precisely known diameter as a length reference. The volume of the sample was obtained from the measured diameter and fitting of spherical harmonic functions to the circumference.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Wilke SK, Alderman OLG, Benmore CJ, Neuefeind J, Weber R. "Octahedral oxide glass network in ambient pressure neodymium titanate." Sci. Rep. 2022 May 18;12:8258. https://doi.org/10.1038/s41598-022-12342-x , May-2022